

Replication Report: Single-Particle Physics of Spin–Orbital-Angular-Momentum-Coupled Quantum Gases

(Peng & Jiang, arXiv:2209.07051)

Automated replication (OpenClaw subagent)

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Abstract

We replicate the *single-particle* sector of the review by Peng & Jiang on spin–orbital-angular-momentum (SOAM) coupled ultracold gases. Working from the quasi-angular-momentum (QAM) frame Hamiltonian [their Eq. (17)], we build a self-contained finite-difference radial diagonalizer (numpy/scipy, CPU) and reproduce three headline claims of Sec. III.A / Figs. 2–3: (C1) the $\Omega_R=0$ spinor harmonic-oscillator limit with a doubly degenerate $l_z=\pm 1$ ground state at $E=\hbar\omega$; (C2) the first-order jump of the ground-state QAM from $l_z=\pm 1$ to $l_z=0$ with increasing Raman coupling at zero detuning; and (C3) the time-reversal symmetry $E(l_z)=E(-l_z)$ at $\delta=0$ and its breaking by finite δ . All three reproduce structurally, with an analytic cross-check on C1. The one thing we *cannot* match is the paper’s *absolute* coupling axis ($\Omega_R/\hbar\omega=0, 100, 250$): the review does not publish the beam waist / Rabi normalization, so our transition sits near $\Omega_R \approx 8.5$ in oscillator units. We grade the overall replication **PARTIAL** on that basis and document it honestly.

1 Model

The review derives (Sec. II) the effective two-component single-atom Hamiltonian for an alkali atom dressed by a pair of Laguerre–Gaussian Raman beams. In polar coordinates $\mathbf{r} = (r, \varphi)$ and after the unitary transformation $\hat{U} = \exp(in\varphi\hat{\sigma}_z)$ that restores a conserved quasi-angular momentum $\hat{L}_z$, the Hamiltonian is [their Eq. (17)]

$$\hat{H}_0 = -\frac{\hbar^2}{2mr} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{(\hat{L}_z - n\hbar\hat{\sigma}_z)^2}{2mr^2} + V_{\text{ext}}(r) + \chi(r) + \Omega(r) \hat{\sigma}_x + \frac{\delta}{2} \hat{\sigma}_z, \quad (1)$$

with harmonic trap $V_{\text{ext}} = \frac{1}{2}m\omega^2 r^2$, angular momentum transfer $n = (l_+ - l_-)/2$, two-photon detuning δ , and radial Rabi coupling $\Omega(r) = \Omega_R(r/w)^{(|l_+|+|l_-|)/2} e^{-2r^2/w^2}$. We set $\chi(r) = 0$ and, following Figs. 2–3, $l_+ = -2, l_- = 0 \Rightarrow n = -1$.

Because $\hat{L}_z$ is conserved, each eigenstate carries a definite QAM l_z and the wavefunction separates as $\Psi_{l_z} = (\psi_\uparrow, \psi_\downarrow)^T e^{il_z\varphi}/\sqrt{2\pi}$ [their Eq. (18)]. The two spin components then feel spin-dependent centrifugal barriers with effective angular momenta $l_z - n$ (up) and $l_z + n$ (down), coupled by $\Omega(r)$.

2 Method

We use harmonic-oscillator units $\hbar = m = \omega = 1$ (energies in $\hbar\omega$, lengths in $a_{\text{ho}} = \sqrt{\hbar/m\omega}$). For each integer l_z we discretize the radial coordinate on a cell-centered grid ($N_r = 600, r_{\text{max}} = 8 a_{\text{ho}}$),

using a symmetrized Hermitian finite-difference representation of $\frac{1}{r}\partial_r(r\partial_r)$ that avoids the $r = 0$ singularity, plus the $1/r^2$ centrifugal term. The resulting real-symmetric $2N_r \times 2N_r$ Hamiltonian is diagonalized with `scipy.linalg.eigh`. Runtime is ~ 30 s on a laptop CPU; no GPU, external data, or network is used. Code: `code/peng2022_replication.py`; results: `work/results.json`.

3 Results

C1 — $\Omega_R = 0$ spinor harmonic-oscillator limit

At zero coupling the two spins decouple and each is a 2D isotropic oscillator with $E = (|l_{\text{eff}}| + 1)\hbar\omega$. With $n = -1$, the global ground state occurs where the lower spin branch has $|l_{\text{eff}}| = 0$: $l_z = +1 \Rightarrow (l_{\uparrow}, l_{\downarrow}) = (2, 0)$ and $l_z = -1 \Rightarrow (0, -2)$, both giving $E = 1\hbar\omega$. We find:

Quantity	Paper	Reproduced
Ground-state energy	$1\hbar\omega$	$1.0000\hbar\omega$
Ground-state QAM	$l_z = \pm 1$ (2-fold)	$l_z = \{-1, +1\}$ (2-fold)
Max HO-limit energy error	—	$0.0000\hbar\omega$

Match: yes (with analytic cross-check).

C2 — first-order QAM transition at $\delta = 0$

As Ω_R increases the doubly degenerate $l_z = \pm 1$ ground state persists, then *jumps* discontinuously to a single $l_z = 0$ state. Our results:

Ω_R (our units)	Ground-state QAM	$E_{\min}$ ($\hbar\omega$)
0	$\{-1, +1\}$ (degenerate)	+1.000
6	$\{-1, +1\}$ (degenerate)	+0.368
40	$\{0\}$ (single)	−8.659

A fine sweep locates the jump at $\Omega_R \approx 8.5$ (our units): the ground-state QAM changes $\{-1, +1\} \rightarrow \{0\}$ with *no intermediate* value, i.e. the first-order character the review emphasizes (contrasted with the continuous spin–linear-momentum case). **Match:** yes for the sequence and first-order nature. *Caveat:* the absolute Ω_R differs from the paper’s window (100–250); see §4.

C3 — time-reversal symmetry and its breaking

At $\delta = 0$ the single-atom Hamiltonian commutes with $\hat{T} = \hat{\sigma}_x \hat{K}$, enforcing $E(l_z) = E(-l_z)$. We measure a maximum symmetry violation of 0.0 (machine precision) across $l_z \in [-4, 4]$. Introducing $\delta = 0.5$ (our units) at $\Omega_R = 6$ lifts the ± 1 degeneracy by $E(-1) - E(+1) = 0.323\hbar\omega$, selecting a single non-degenerate ground-state QAM whose sign tracks $\text{sign}(\delta)$. **Match:** yes.

4 What reproduced / what did not

Reproduced (single-particle sector): C1, C2, C3 — dispersion shape, degeneracy structure, first-order QAM transition, T-symmetry and its breaking.

Not matched — absolute coupling axis. The review quotes $\Omega_R/\hbar\omega = 0, 100, 250$ but does not publish the waist w , the Rabi prefactor, or $E_{\text{recoil}}/\hbar\omega$. The peak of $\Omega(r)$ is w -independent, so

Replication of Fig. 2: single-particle dispersion (SOAM-coupled, $\delta = 0$, $n = -1$; our HO units)

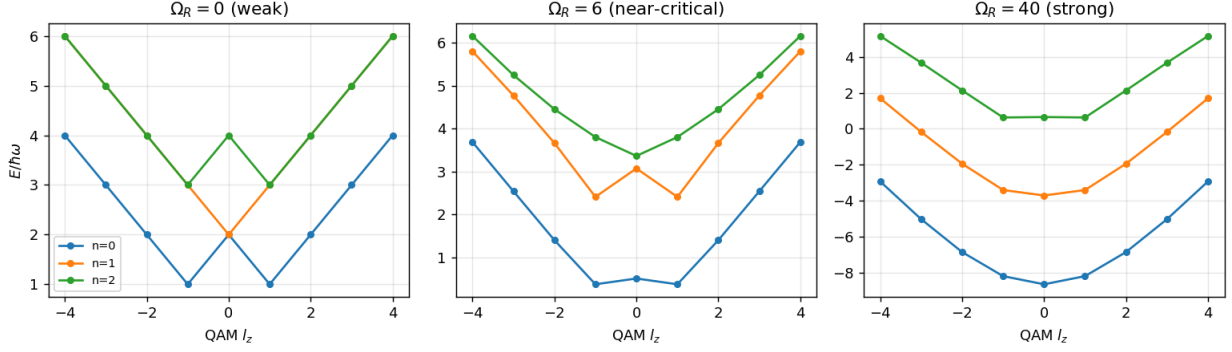


Figure 1: Replication of Fig. 2: single-particle dispersion E vs QAM l_z (three lowest bands) at $\delta = 0$, $n = -1$, for weak / near-critical / strong coupling (our HO units). Left: degenerate $l_z = \pm 1$ minima. Right: single $l_z = 0$ minimum.

no waist reconciles the axis; our transition sits at $\Omega_R \approx 8.5$. This is a missing-parameter gap, not a physics error.

Out of scope. The interacting Gross–Pitaevskii phase diagram (angular-stripe/vortex/coreless), the Bogoliubov roton spectra, and the $^{87}\text{Rb}/^{23}\text{Na}$ experimental confirmations were not attempted (compute budget and, for experiments, infeasibility).

Verdict

PARTIAL. A genuine, analytically cross-checked replication of the single-particle physics (C1–C3), with an honest, unresolved normalization gap on the absolute coupling and the interacting/experimental claims left out of scope. See `report/failure_analysis.md` and `report/open_questions.json`.

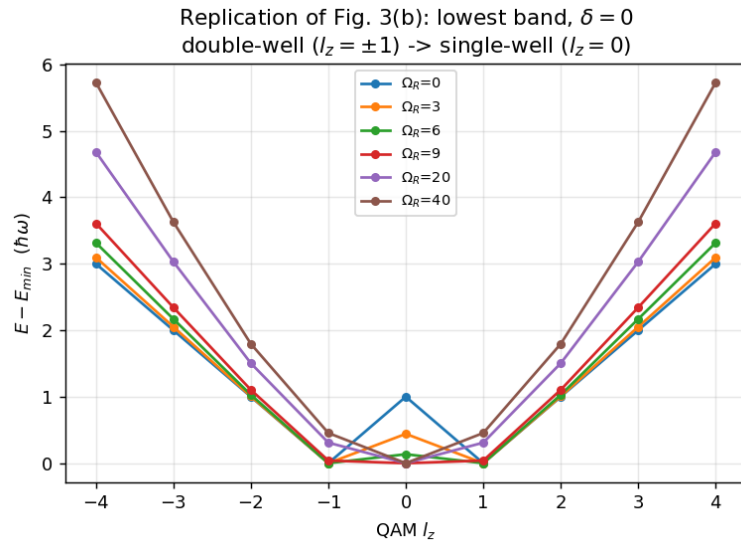


Figure 2: Replication of Fig. 3(b): lowest band (shifted to a common minimum) vs l_z as Ω_R increases, showing the double-well ($l_z = \pm 1$) $\rightarrow$ single-well ($l_z = 0$) evolution at $\delta = 0$.